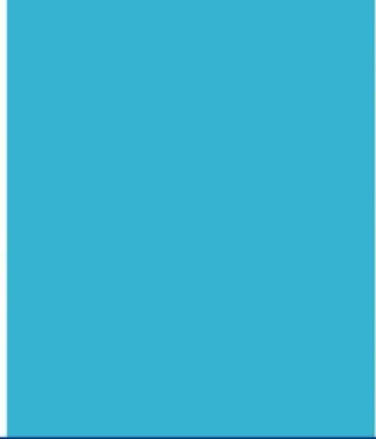




มหาวิทยาลัยมหิดล



Mahidol University

Wisdom of the Land

Miscellaneous bacteria

MTMI 203 Medical Bacteriology and Mycology I

Assoc. Prof. Warawan Eiamphungporn, Ph.D.

11th March 2025



Miscellaneous bacteria

The bacteria stain poorly with conventional dyes
and do not grow on common laboratory media.
Some are uncultivable and do not grow on any
artificial medium.



- **Family Treponemataceae**
 - *Treponema*
- **Family Borreliaceae**
 - *Borrelia*
- **Family Leptospiraceae**
 - *Leptospira*
- **Family Rickettsiaceae**
 - *Rickettsia* and *Orientia*
- **Family Chlamydiaceae**
 - *Chlamydia*
- **Family Mycoplasmataceae**
 - *Mycoplasma* and *Ureaplasma*
- **Family Legionellaceae**
 - *Legionella pneumophila*

Outline

- Morphology and cultural characteristics
- Clinical significance
- Epidemiology
- Laboratory diagnosis
- Treatment, prevention and control



Order Spirochaetales

- Family Treponemataceae
 - 1 medically important genus
 - *Treponema*
- Family Borreliaceae
 - 1 medically important genus
 - *Borrelia*
- Family Leptospiraceae
 - 1 medically important genus
 - *Leptospira*

→ ฟิลิเคิลซว



Family ^①Treponemataceae

- Morphology and cultural characteristics:
 - They are slender unicellular helical or spiral rods.
 - They contain flagella and are actively motile.
 - ^{จับย้อมสีไม่ได้} They are Gram negative but most of the strains do not take up stain easily by Gram staining or Giemsa staining.
 - * They can easily seen on dark-field microscopy. *
 - Pathogenic treponemes cannot be cultivated in laboratory media and are maintained by subculture in susceptible animals. *



Treponema

- Treponemes are **obligate intracellular parasites**.
- They are **microaerophiles** or **anaerobes**. ต้องการ O_2 น้อย หรือ ไม่ต้องการ O_2
- The pathogenic **treponemes for human** including

① *Treponema pallidum*:

- subspecies pallidum → ¹⁾ **Syphilis** ^{*} (venereal or sexually transmitted disease)
- subspecies endemicum → ²⁾ **Bejel** ^{พบบ่อยพบเฉพาะถิ่นออกภพ} (endemic syphilis)
- subspecies pertenue → ³⁾ **Yaws**

ไม่ติดต่อ
non-venereal
diseases

② *Treponema carateum*: Pinta

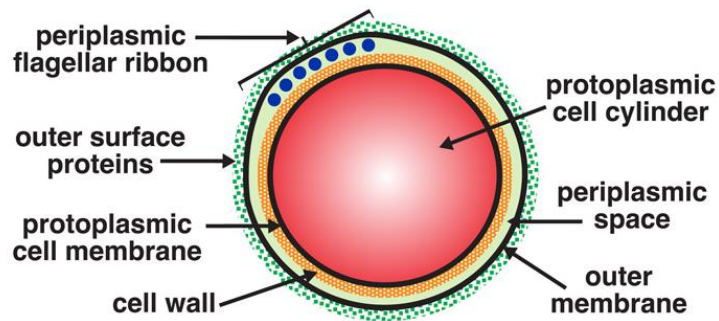
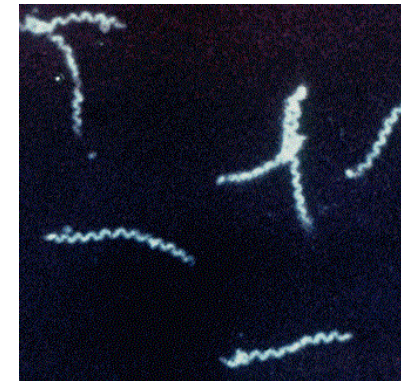
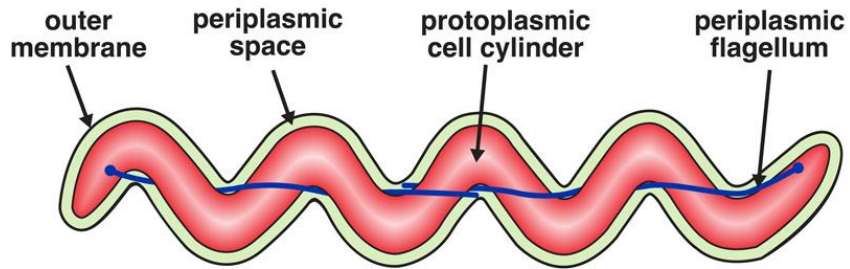


Treponema pallidum subsp. *pallidum*

- It is very thin and tightly coiled helical rod with pointed, straight end. *
- It contains 3 periplasmic flagella at each end, wrapping around its body.
- It can produce outer membrane proteins (adhesion) and hyaluronidase (invasion).
①
ชั้นเยื่อหุ้ม mucus membrane การเกาะจับกับเนื้อเยื่อ
- It has the ability to cross intact mucous membranes and placenta, disseminate throughout the body and infect almost any organ system.
②
- It is sensitive to heat, drying and disinfectants.
ร้อน แห้ง ยาฆ่าเชื้อโรค



Treponema pallidum subsp. *pallidum*





Phases of syphilis

- **Primary phase syphilis** แผลหนอง
 - one or more skin lesions (chancre) at the site of infection แผลที่รอยโรค / แผลริมแท้
 - In most cases, a painless lymphadenopathy ต่อมน้ำเหลืองโต
- **Secondary phase syphilis** แผลหนอง
 - Usually start with flu-like syndrome following by skin lesions over the body surface including palms and soles เริ่มแรก
- **Latent syphilis** ดูภายนอกไม่รู้ว่าป่วย แต่อาจได้ผ่านเลือด
 - During this time, the body will harbor the disease without symptoms. This phase can last for several years.
- **Tertiary phase syphilis**
 - **!** Granulomatous lesions (gummas) may be found in bone, skin and other tissues and degenerative changes in CNS (neurosyphilis). ทำลายเนื้อเยื่อแทบทุกอวัยวะ
 - Tissue destruction in virtually all organs can occur. → หนอง / ฝี / ฝีในอวัยวะ

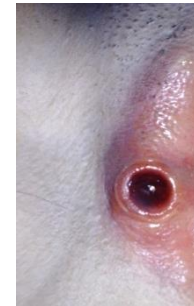


! Congenital syphilis !

- Treponemes can be transmitted from an infected mother to her fetus by crossing the placenta.*
- Congenital syphilis affects many body systems and is severe and mutilating.
- Early-onset^{กล่าว 2} congenital syphilis resembles secondary syphilis in adults.
- Late-onset^{กล่าว 3} congenital syphilis corresponding to tertiary syphilis in adults.



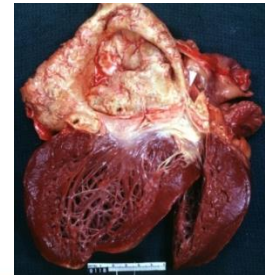
Primary phase syphilis



Secondary phase syphilis



Tertiary phase syphilis





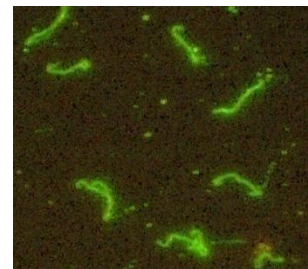
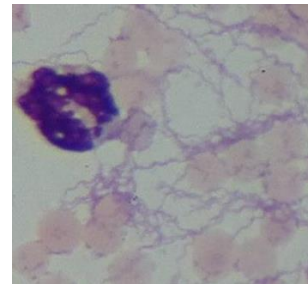
Epidemiology

- **Humans are the only natural host.**!
พู่กัน Syphilis ไม่สามารถเพิ่มเชื้อ ในร่างกายของคนที่มี HIV เชื้อจะรอโรคที่จักษุจะเกิด
- Syphilis patients are at increased risk for transmitting and acquiring HIV when genital lesions are present.
- Thus, this disease remains a serious problem in sexually active population.
- During the early stages, the patient becomes bacteremic.*
- Syphilis is contagious during its primary and secondary stages.
สามารถติดต่อได้ตอน 1st และ 2nd



Laboratory diagnosis

- Microscopy
 - Treponema are too thin to be seen under light microscope after staining specimen with Gram or Giemsa stain.
 - Motile activity can be seen under dark-field *
 - ! The most useful test is direct fluorescent antibody (DFA) test. !
 - Fluorescein-labeled monoclonal antibodies can be used to examine the specimens under fluorescent microscope.

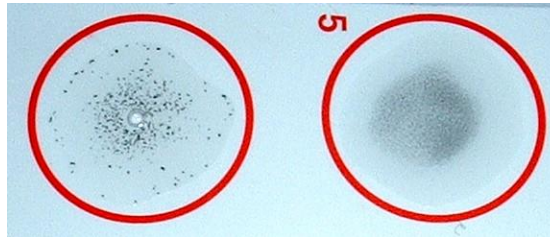




Laboratory diagnosis

- Antibody detection

- Nontreponemal tests are not specific (using cardiolipin as an antigen) but rapid and inexpensive; Venereal Disease Research Laboratory (VDRL) test and Rapid plasma reagin (RPR) test



- Treponemal test is specific; *Treponema pallidum* particle agglutination (TP-PA) test





Treatment, prevention and control

- **Penicillin** is the drug of choice.
- ~~Syphilis vaccine is not available~~
- Prevention can be done by safe-sex practices.
- Contact and treatment of partner who has been documented for the disease.
- Screening of pregnancy woman, and donated blood, for *T. pallidum* infection is the current practices for controlling syphilis.



Family Borrieriaceae

- Morphology and cultural characteristics:
 - *Borrelia* are Gram-negative **spirochetes**.
 - They can be well stained by Giemsa and Wright stain but weakly stain by Gram. They can be visualized by bright-field microscopy!
 - They contain 15-20 periplasmic flagella, which responsible for !twisting motility.!
 - They are microaerophilic.
 - Many species can be cultured in vitro in enriched, serum containing, media.
แอลกอฮอล์ที่ฆ่าเชื้อเพิ่ม & serum อยู่ด้วย



Borrelia

- *Borrelia* are the causative agents of Lyme disease and relapsing fever.
- **Lyme disease** is caused by *B. burgdorferi*, *B. garinii* and *B. afzelii* which are transmitted by hard ticks. !
- **Relapsing fever** can be divided into two types:
 - **Epidemic relapsing fever** is caused by *B. recurrentis* and is spread by human louse. !
 - **Endemic relapsing fever** is caused by as many as 15 species of borreliae and is spread by soft ticks. !



Lyme disease

พิกัดเส้นแวงการแพทย์

พิกัดเส้นแวงการแพทย์

- After tick bite 3-30 days, the erythema migrans developed and central necrosis may be seen but แต่ * they are not pathognomonic sign. ไม่ใช่ Sign เฉพาะโรค
- The other initial symptoms include malaise, severe fatigue, headache, fever, musculoskeletal pain, lymphadenopathy.
- In untreated patients, the disease develop * systemic symptoms, arthritis, arthralgia, cardiac dysfunction and neurologic signs. Arthritis is the ! most common symptom. ! ข้อควรระวัง





สามารถ change Antigen 4u
ผิว cell ใหม่ → Relapse ใหม่

Relapsing fever

- Relapsing fever is a febrile disease characterized by recurrent fever and septicemia separated by non-febrile periods.
- After 1 week of infection, patient develop shaking chills, fever, muscle aches and headache.
- These symptoms correspond with bacteremia and resolve within 7 days. However, these symptoms return after non-febrile period.
- For immune response, when specific antibodies are formed and the bacteria is rapidly cleared from circulation.
- The relapsing cycles correlate with antigenic phase variation of outer membrane proteins.



Borrelia



Lyme disease

(hard tick-borne)

- *B. burgdorferi*
- *B. garinii*
- *B. afzelii*



Relapsing Fever

Epidemic

(louse-borne)

- *B. recurrentis*



Endemic

(soft tick-borne)

- other *Borrelia*



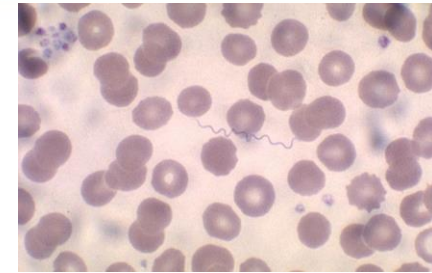


Epidemiology

- **Lyme disease**
 - Worldwide distribution
 - Seasonal incidence; most cases in USA occur in late spring and early summer.
 - Reservoir; mice, deer, hard tick
- **Epidemic** relapsing fever
 - Occurs in Northeast Africa
 - Reservoir; human
- **Endemic** relapsing fever
 - Worldwide distribution and western part of USA
 - Reservoir; rodents, small mammals, soft tick



Laboratory diagnosis



- Microscopy

- Relapsing fever: Blood specimen with Giemsa or Wright stain
- Lyme disease: Blood specimen not recommended*

- Culture

- Borreliae can be cultivated using Barbour-Stoenner-Kelly medium.

- Nucleic acid-based test (PCR)

- Serology *

- It is a diagnostic test for Lyme disease but it is not useful for relapsing fever because the bacteria undergo antigenic phase variation.



Treatment, prevention and control

- Relapsing fever; Doxycycline is the primary treatment.
- Lyme disease; Amoxicillin, doxycycline and cefuroxime are the first-line antimicrobials for treatment.
- **! Vaccine is not available. !**
- Arthropod control and rodent control
- Improve hygiene



Family Leptospiraceae

- At least 64 species and 250 serovars of *Leptospira* have been recognized:
 - Pathogenic *Leptospira* → *L. interrogans*, *L. noguchii*, *L. alstonii*
 - Intermediate or opportunistic *Leptospira* → *L. broomii*, *L. inadai*
 - Non-pathogenic *Leptospira* → *L. biflexa*, *L. idonii*, *L. meyeri*
- Each specie can be divided into many serovars.
- It is a zoonotic disease.
- The most common reservoirs are rodents and other small mammals, particularly rats.
- *Humans are dead-end accidental hosts.*



Leptospira



- Morphology and cultural characteristics:
 - It is thin, coiled spirochetes with a hook at one or both ends.
 - It has 2 periplasmic flagella, which responsible for highly motile.
 - It is an obligate aerobe with optimum growth at 28-30 °C.
 - It can be cultured in artificial media, such as Fletcher or Ellinghausen-McCullough-Johnson-Harris (EMJH) semisolid.!



Leptospira

- It is an etiologic agent of **leptospirosis**.
- They can penetrate intact mucous membrane or skin through small cuts or abrasions. *မှန်ကန်စွာ*
- Infection can induce immune response can cause glomerulonephritis.
- In most cases, leptospires infections are clinically inapparent.*
However, symptoms usually develop 1-2 weeks after infection.
- The bacteria can be found in blood and CSF during early and in urine during late stages.
- Aseptic meningitis may be found up to 25% of all cases.



Leptospira

- Pathogenic leptospire can cause two phases of infection.
 - Subclinical infection: influenza-like febrile illness with fever, myalgia, chills, headache, vomiting, or diarrhea
 - Severe infection (Weil syndrome): ^{ปวดหัวรุนแรง} sudden headache, ^{กล้ามเนื้ออ่อนแรง} myalgia, ^{เยื่อตาแดง} conjunctival suffusion, vascular collapse, ^{เกล็ดเลือดต่ำ} thrombocytopenia, hemorrhage, ^{4 ทกพ} jaundice, hepatic and renal failure, myocarditis and death.





Epidemiology

- The bacteria colonize in renal tubules of reservoir hosts and are shed largely in the urine.
- Bacteria can stay in stream or standing water, and moist soil for up to 6 weeks.
- Humans are infected through recreational or occupational exposure to contaminated water or through handling infected animal tissues.
- * Person-to-person spread has not been documented. *

ไม่มีคนติดต่อ: ไม่มีคนติดต่อ



Epidemiology

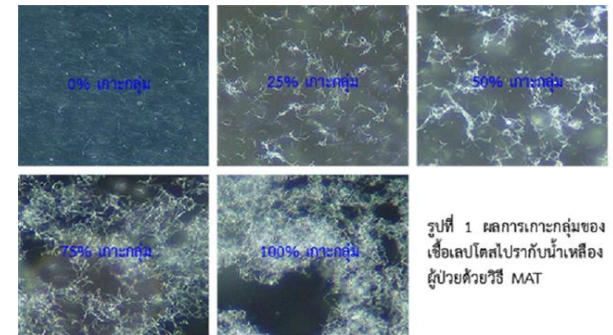
- People at risk are those who expose to water contaminated with animal urine and expose to infected animals (farmers, meat handler, and veterinarians).
- Infection has worldwide distribution but more common in tropical area, especially during warm months.





Laboratory diagnosis

- Microscopy
 - Dark-field microscopy are insensitive. !
 - DFA technique can be used. !
- Culture
 - It can be cultured in special media (2 weeks-4 months).
 - Positive culture can be seen under dark-field microscope.
- Nucleic acid-based test (PCR)
 - PCR is more sensitive than culture.
- Serology !
 - Most cases are diagnosed by serology (ELISA)
 - The reference method is microscopic agglutination test (MAT).





Treatment, prevention and control

- Penicillin is the primary treatment.
- Doxycycline and ceftriazone are alternatives.
- Vaccination in livestock and pets has proved successful in reducing disease.
- Rodent and water control are also effective in eliminating the disease.

↓
ควบคุม สัตว์ฟันแทะและน้ำ
ดื่ม สดในภาชนะ ปิดมิดชิด
โรค



Family Rickettsiaceae

- 2 medically important genera:
 - *Rickettsia* are recognized at least 30 species of and can be divided into 3 groups.
 - Spotted fever group; *R. rickettsii* (Rocky mountain spotted fever)
 - Typhus group; *R. prowazekii* (Epidemic or louse-borne typhus) and *R. typhi* (Endemic or murine typhus)
 - Transitional group *R. akari* (Rickettsialpox)
 - *Orientia tsutsugamushi* causes scrub typhus. *ไท้รอกสุมุชิ*



Family Rickettsiaceae

- Morphology and cultural characteristics:
 - All are **obligate intracellular parasites** ^{*} that **grow only** in the **cytoplasm of eukaryotic cells**.
 - They are **small, aerobic, non-motile** **pleomorphic rods** [!] or coccobacilli.
 - **Gram stain is poorly**, but appears to be Gram negative.
 - [!] **Stain readily with Giemsa stain** ^{*}
 - **They are grown in embryonated eggs or tissue culture.**



Rickettsia and Orientia

- They are maintained in reservoir hosts (primarily rodents) and their arthropod vectors (e.g., tick, mite, louse, flea).!
- ! Humans are accidental hosts except *R. prowazekii*.!
- They enter endothelial cells by attaching to cell surface receptors and stimulating phagocytosis.
- After engulfment, they ^①degrade the phagosome and ^②replicate in the cytoplasm of the host cells.
- Typhus rickettsiae differ from other groups in that they cause ! cell lysis, thereby releasing the bacteria.
- Other rickettsiae pass directly through an uninjured cell.

↳ ไม่เจอ กับ immune cell



Rickettsia and Orientia

- Common symptoms typically develop within 1-2 weeks of infection including fever, headache, malaise, rash, nausea and vomiting.
- Many rickettsioses are accompanied by a maculopapular, vesicular or petechial rash or eschar at the site of the tick bite.



Petechial rash



Eschar



Typhus rash



Epidemiology

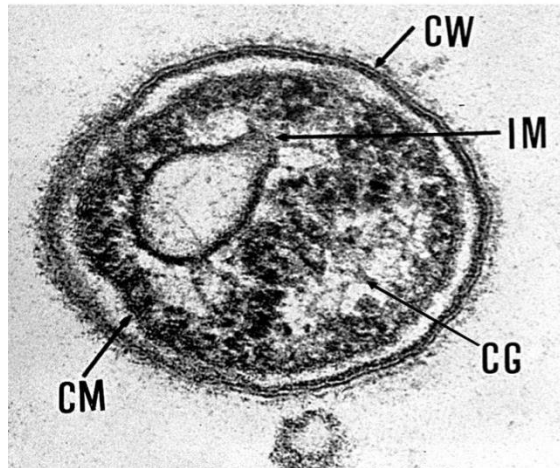
- *R. rickettsii* and *R. akari*
 - USA ,Canada, Central and South America
- *R. prowazekii* and *R. typhi*
 - Worldwide distribution, especially in tropical and subtropical regions.
- *O. tsutsugamushi*
 - Japan, eastern Asia, southeast Asia, northern Australia, west and southwest Pacific



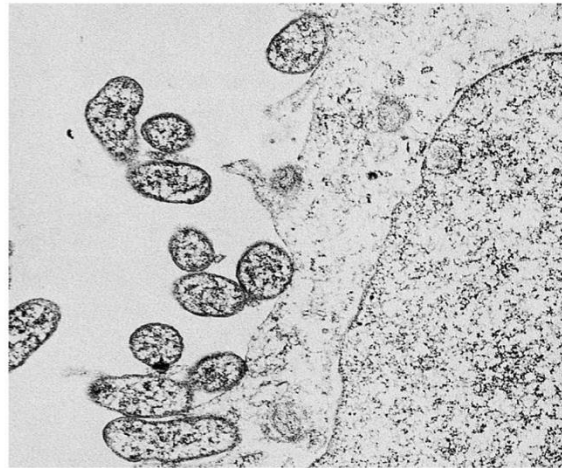
Rickettsia

- Transmission

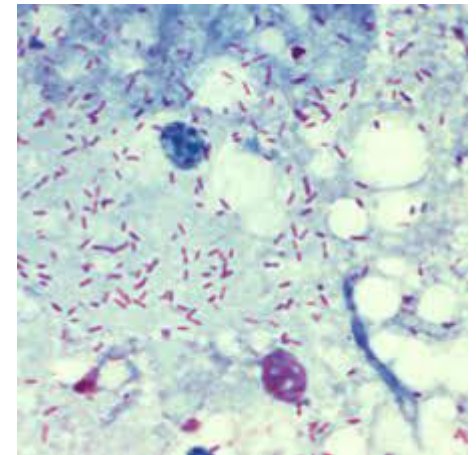
- ① – *R. rickettsii* → transmitted by **hard ticks** (dog and wood ticks) !
- ② – *R. prowazekii* → transmitted by **human lice** !
- ③ – *R. typhi* → transmitted by **fleas** !
- ④ – *R. akari* → transmitted by **mouse mites** !
- ⑤ – *O. tsutsugamushi* → **chiggers** (mite larvae) !



(a) Cell morphology



(b) Budding from a cell



(c) Gimenez stain





Laboratory diagnosis

- Microscopy
 - Detection of *Rickettsia* in tissue by Giemsa or Gimenez stains or DFA test
- Serology
 - ELISA
 - Immunofluorescence (IFA) test (reference method)
 - Weil-Felix test (insensitive and nonspecific test)
- Nucleic acid-based tests (PCR)
- Culture (tissue culture or embryonated eggs)!



Treatment, prevention and control

- The drug of choice is doxycycline.!
- People should avoid tick-infected areas, wear protective clothing and use effective insecticides.
- Controlled through improvements in living conditions.
- ! Inactivated vaccine is available for high-risk populations.



Family Chlamydiaceae

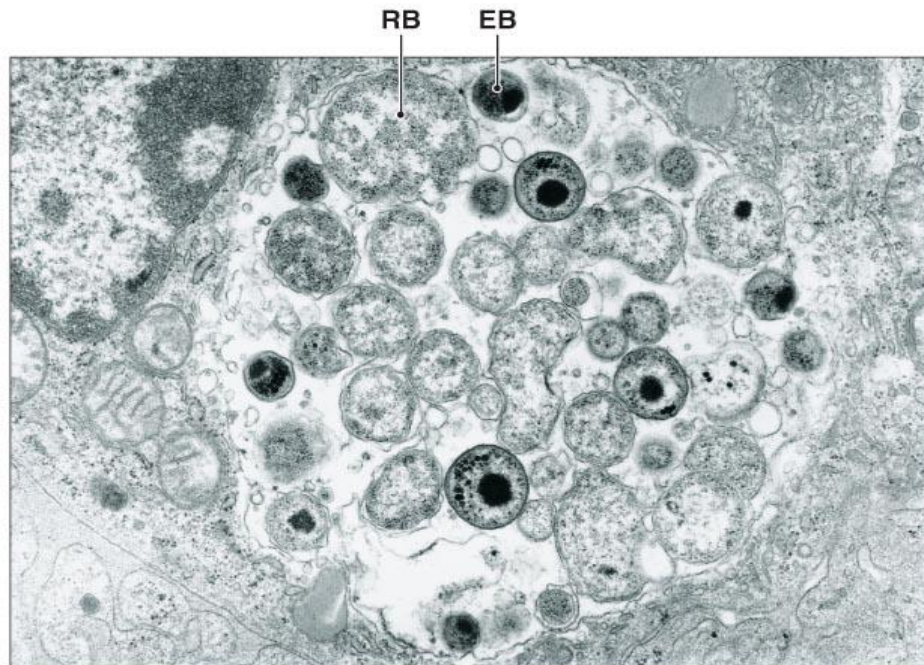
- 1 clinically important genus:
 - Chlamydia with 3 species responsible for human diseases
 - ① • Chlamydia trachomatis
 - ② • Chlamydia pneumoniae
 - ③ • Chlamydia psittaci
 - Other species of Chlamydia exist, but they are rarely isolated from humans.



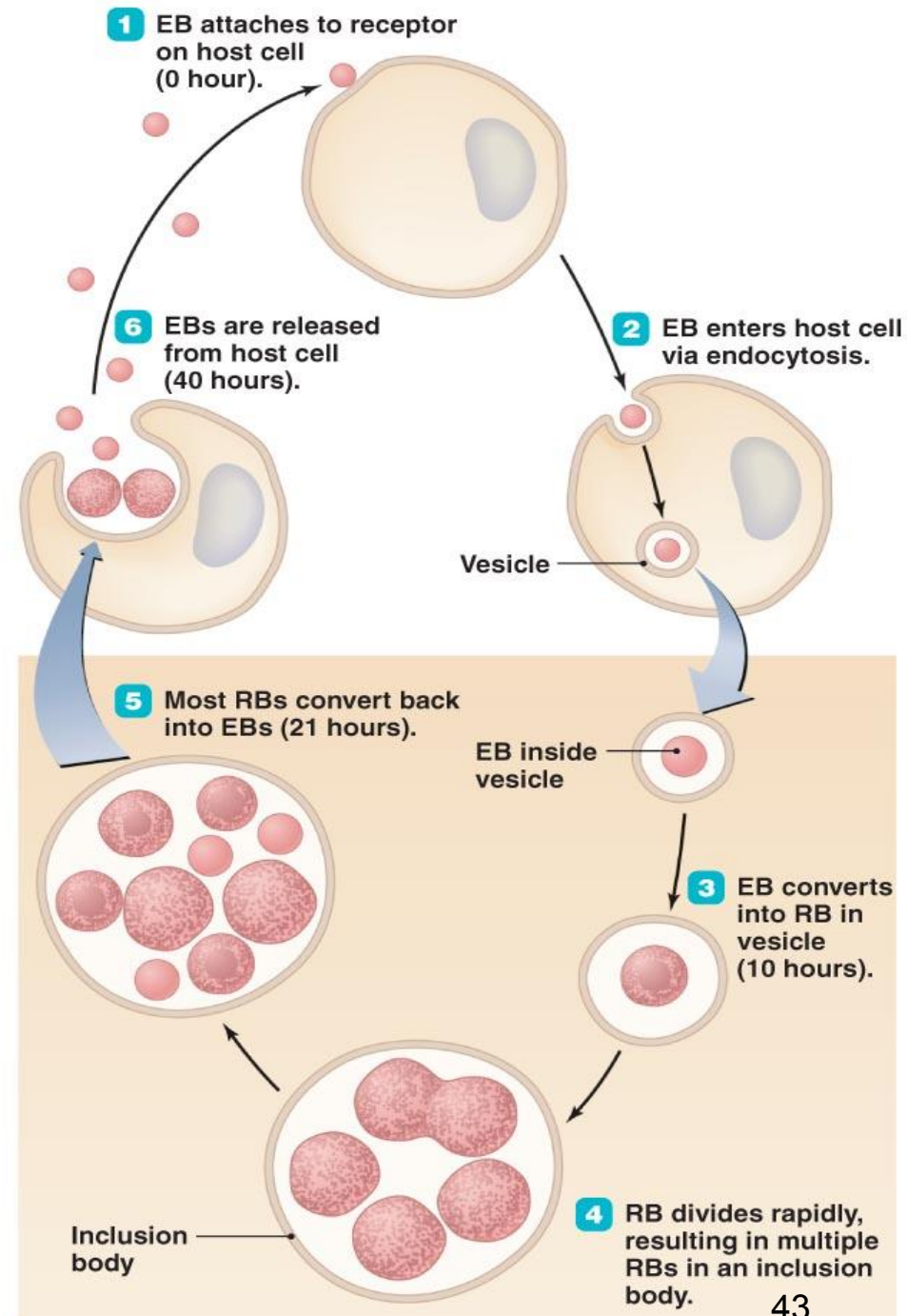
Family Chlamydiaceae

- Morphology and cultural characteristics:
 - Chlamydiales are small Gram-negative bacteria.
 - They are the **obligate intracellular parasites** which **grow** and multiply only within the vesicles of host cells.
 - They have a unique developmental cycle involving **two forms**:
 - 1) **Elementary bodies (EBs)**: small metabolically inactive, extracellular infectious forms released by the infectious cell.
 - 2) **Reticulate bodies (RBs)**: non-infectious, actively dividing forms, **grow** within host cell vacuoles.
 - **A Giemsa stain can be used** to visualize the inclusions in tissues.

Development forms and life cycle of *Chlamydia*



(a)



(b)



Chlamydia trachomatis

- Infections restricted to humans. !
- Species cause disease are subdivided into two biovars:
 - 1) Trachoma
 - 2) Lymphogranuloma venereum (LGV)
- The biovars have been further divided into serovars (A-L).
- The serovars determine clinical manifestation.
- Clinical significance; Infected nonciliated columnar, cuboidal and transitional epithelium cells.



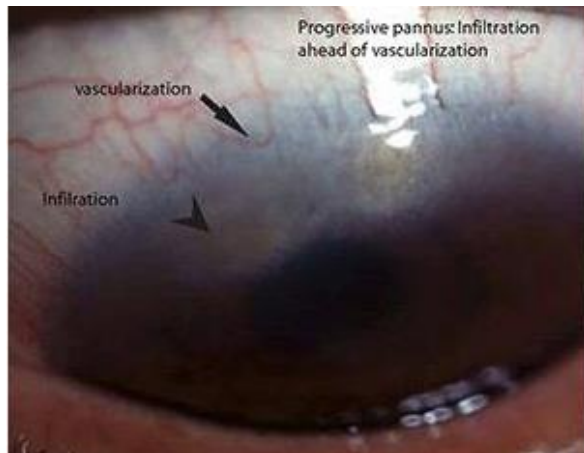
Chlamydia trachomatis

- The clinical manifestation:
 - ^{โรคตาขาว} Trachoma (A-C)
 - Urogenital infections (D-K); ^{หนองในเทียม} non-gonococcal urethritis, proctitis, epididymitis, cervicitis, salpingitis and pelvic inflammatory disease (PID)
 - Adult inclusion conjunctivitis (D-K)
 - Neonatal conjunctivitis (D-K)
 - Infant pneumonia (D-K)
 - ^{ฝีหนอง} LGV (L1-L3)



Primary ocular chlamydial infection

Trachoma



Neonatal conjunctivitis 46



Epidemiology

- Common sexually transmitted bacterial infection. !
- Most common cause of **nongonococcal urethritis (NGU).** !
- Most genital tract infections in women are asymptomatic. *
- Dual infection with *N. gonorrhoeae* is common. ^{ติดต่อร่วม}
- Ocular trachoma primarily in Africa, South Asia, Middle East and South America
- LGV highly prevalence in Africa, Asia and South America



Chlamydia pneumoniae

- ^{ไม่รู้อะไรเลย} No animal reservoir or vector is known.*
- Infection can range from ^① asymptomatic colonization to ^② severe bronchopneumonia.
- Most infections do not require hospitalization.
- Severe cases can resemble primary atypical (walking) pneumonia (fever, chills, dry cough, headache, sore throat, nausea, and vomiting) caused by *M. pneumoniae*.
- ^{!!} Prevention is difficult because *C. pneumoniae* is ubiquitous. ^{พบได้ทั่วไป}



Chlamydia psittaci

- Naturally infects avian species and non-primate animals causing mild to severe illness.
- ! It is a zoonotic disease. *
- In man, it causes ^{ไข้นกแก้ว} **psittacosis (ornithosis)** and is acquired by inhalation of infected dust from droppings or sneezed aerosol droplets.
 - Infection can range from asymptomatic or mild to severe disease.
 - Symptoms include fever, chills, muscle aches, and severe headache.
 - Most commonly [!] causes an atypical pneumonia. *

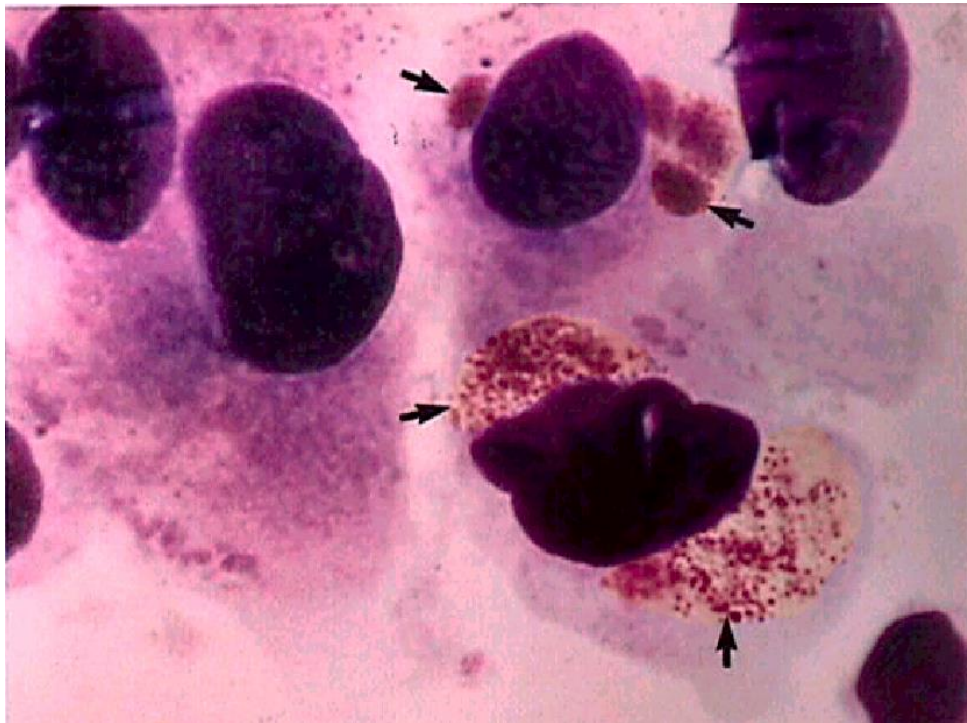


Laboratory diagnosis

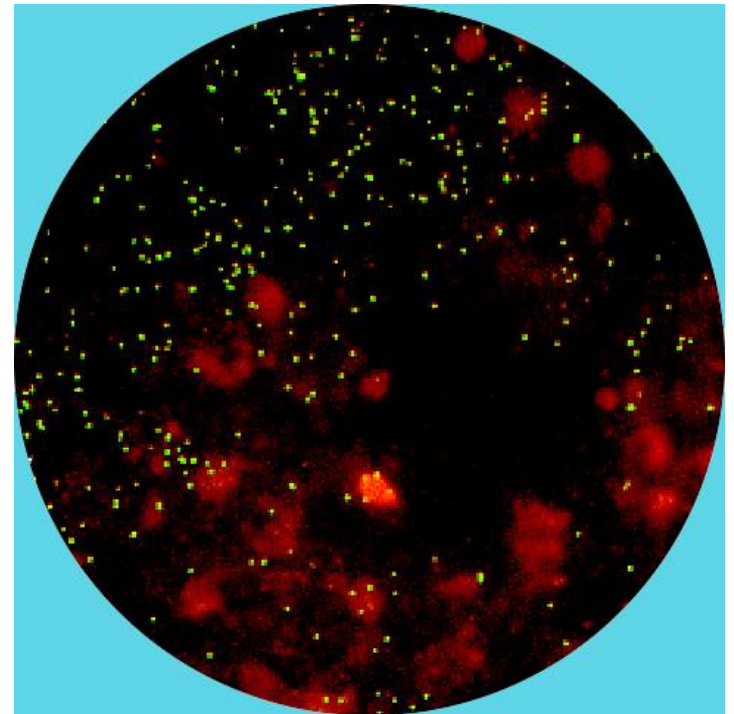
- Microscopy^①
 - Stain tissues with Giemsa or use a DFA technique^②.
- Antigen/antibody detection
 - ELISA
 - A complement fixation
 - Immunofluorescence (IFA) test
- Culture (Isolation)
 - ! Gold standard, highly specific but relatively insensitive
- Nucleic acid detection PCR
 - ! The most sensitive and specific (90-98% sensitivity)



Chlamydia in tissues



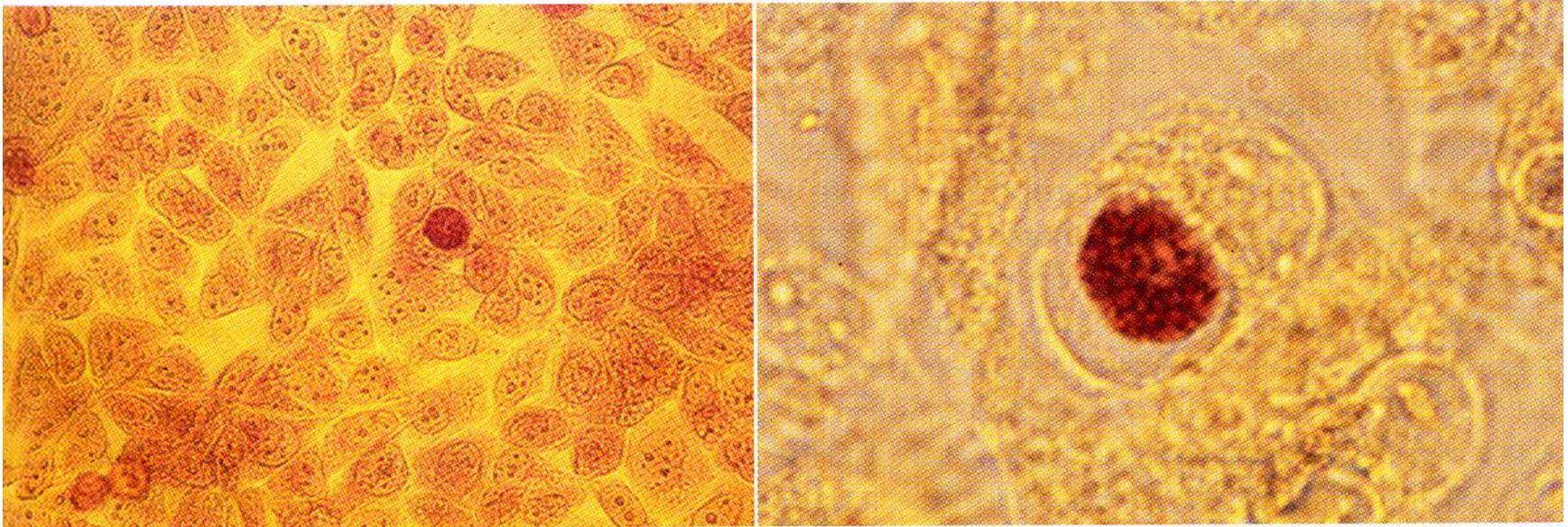
Giemsa staining- EBs



DFA staining- EBs



Chlamydia trachomatis in tissues



Iodine staining- *C. trachomatis* **inclusion bodies** *



Treatment, prevention and control

- *C. trachomatis*
 - For trachoma, a single dose of azithromycin is the treatment of choice.
 - Azithromycin is the preferred therapy for non-LGV.
 - Doxycycline is the drug of choice for treating LGV.
 - Safe sex practice and prompt treatment of patient and sexual partners help to control the infection.



Family Mycoplasmataceae

- 2 medically important genera:

- ① – *Mycoplasma*

- ② – *Ureaplasma*

- 4 common clinical isolates:

- *M. pneumoniae*
- *M. genitalium*
- *M. hominis*
- *U. urealyticum*



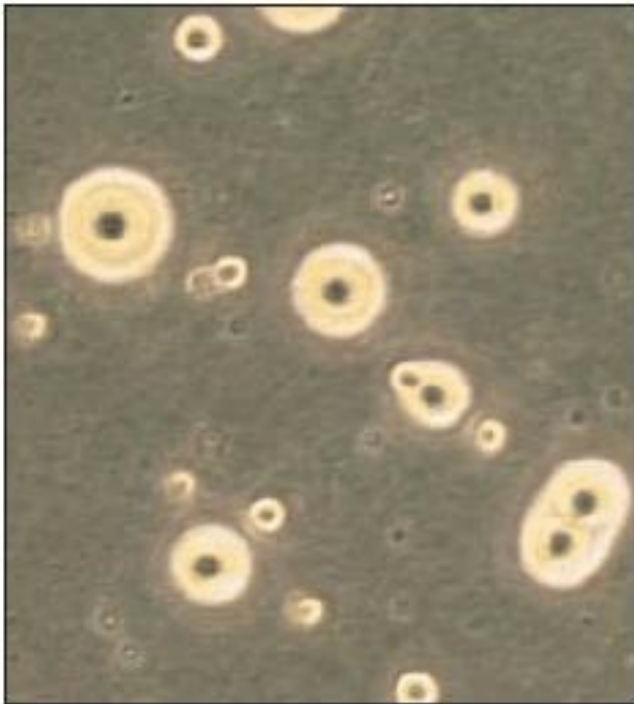
Family Mycoplasmataceae

- Morphology and cultural characteristics:

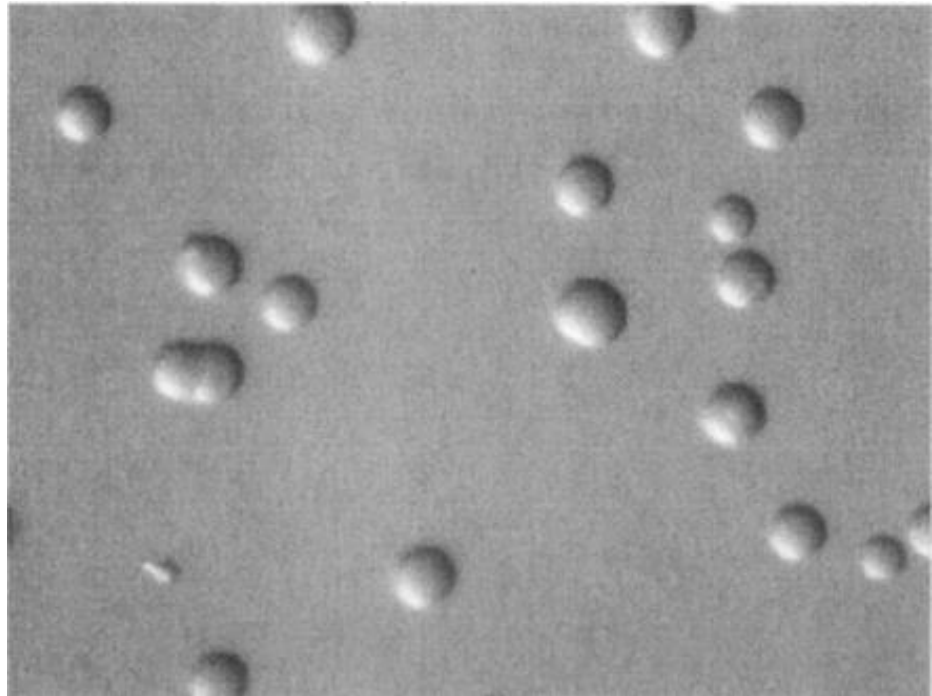
- They are the ^{* Bacteria ที่เล็กที่สุด *} smallest free-living bacteria and highly pleomorphic. !
- They ^① do not have a cell wall and their ^② cell membrane contains sterols that act to prevent osmotic lysis. !
- They do not stain with a Gram stain. *
- They are facultatively anaerobic ^{↑ ใช้ออกซิเจน O₂} except *M. pneumoniae* (strict aerobe). !
- Grow on artificial cell-free media enriched with serum. !
- Typical mycoplasmal colonies resemble fried eggs, except *M. pneumoniae* exhibits spherical colonies.



Mycoplasma colonies



Typical fried egg colonies



Spherical colonies



Mycoplasma and Ureaplasma

- Clinical significance:
 - *M. pneumoniae* is transmitted by inhalation of aerosolized droplet. It adheres to the respiratory epithelium and the most common clinical presentation is **tracheobronchitis**. It is the major cause of primary, **atypical pneumonia (walking pneumonia)**.
 - *M. genitalium* and *U. urealyticum* can cause **nongonococcal urethritis (NGU)** and other urogenital diseases.
 - *M. hominis* has not been reported as a cause of **NGU**. The organism causes **salphingitis, pyelonephristis, pelvic inflammatory disease (PID) or postpartum fever**.



Epidemiology

- *M. pneumoniae* (Eaton agent):
 - It is a strict human pathogen.
 - Worldwide disease with no seasonal incidence
 - Most common in children and young adults between 5-15 years, but all populations susceptible to disease
 - Epidemic disease occurs every 4 to 8 years.
- Genital mycoplasmas:
 - Genitourinary tract colonization with *M. hominis*, *M. genitalium* and *Ureaplasma* spp. is common. ~15% of sexually active men and women are colonized with *M. hominis* and higher proportion with *M. genitalium* and *Ureaplasma* spp.



Laboratory diagnosis

- Microscopy
 - It is not useful because *Mycoplasmas* stain poorly.!
- Culture
 - It is relatively insensitive method since the organisms grow slowly.!
- Nucleic acid-based tests PCR ✓
 - They have high sensitivity and specificity.!
- Serology
 - Antibody-specific tests are available for *M. pneumoniae*.
 - Serologic methods are available for others but are not recommended for routine diagnosis.



Treatment, prevention and control

- Tetracyclines (particularly doxycycline), azithromycin or newer fluoroquinolones are drug of choice for *Mycoplasma* spp.
- *M. pneumoniae* infections are spread by close contact, thus the isolation of infected people could reduce the risk of infection.
- *M. hominis*, *M. genitalium* and *U. urealyticum* infections can be prevented by avoidance of unprotected sexual activity.
- Vaccines have proved ineffective.



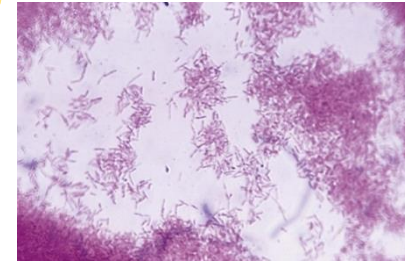
Family Legionellaceae

- 1 medically important genus
 - *Legionella* with 50 species (half of these species can cause disease in human and >90% of infections are *L. pneumophilla* !)
 - *Legionella* was first isolated from American soldiers with severe pneumonia during its outbreak in Philadelphia in 1976.



Legionella pneumophila

- Morphology and cultural characteristics:
 - It is slender, pleomorphic Gram-negative bacilli.
 - It is an **obligate aerobe** in nature and nutritionally **fastidious**! (does not grow on blood agar).
 - The bacteria appear as coccobacilli in tissue specimen but very pleomorphic in artificial media.
 - Bacteria in tissue cannot be stained with common dyes but can be stained with silver stain.!
 - Legionellae are facultative intracellular parasites.!





Legionella pneumophila

- Clinical significance:
 - The bacteria are capable of replication in alveolar macrophages, monocyte, and alveolar epithelial cells (and amoebae in nature).
 - **!** It can prevent phagolysosome fusion, which results in intracellular survival and consequently pneumonia.
 - Human infection is associated with exposure to contaminated aerosols.
 - Symptomatic infections present in one of two forms
 - **!** An influenza-like illness (Pontiac fever)
 - **?** A severe form of pneumonia (legionnaires' disease)
 - ***** It causes a disease referred as atypical pneumoniae.*****



Epidemiology

- Legionellae have worldwide distribution.
- Adults over 40 are at the higher risk for legionnaires disease.
- The bacteria are commonly present in natural water, as well as air conditioning cooling tower, condenser and water systems.
- ^① Person-to-person spread or an ^② animal reservoir has not been demonstrated.!



Laboratory diagnosis

- Microscopy
 - Gram stain is not sensitive enough.
 - ! Silver stain can be used but this method is for tissue not sputum.
 - DFA is low sensitive (25-50%)
- Nucleic acid based test (PCR)
 - Sensitivity is equivalent to culture.
 - Specimens from patients can be used but cultured bacteria yields more sensitivity.



Laboratory diagnosis

- Culture

- * Bacteria can be easily grown in BCYE α (Buffered Charcoal Yeast Extract + 0.1% α -ketoglutaric acid) agar (Gold standard). It grows at 37 °C, 3-5% CO₂ (ground-glass appearance).
- Specie identification can be done by biochemical testing and the ability to fluoresce under long-wave UV light.



- Serology

- ELISA but it is insensit[!]ive and nonspec[!]ific, especially when polyclonal antibodies are used.
- ! Urinary antigen test is highly sensit[!]ive (60-90%).





Treatment, prevention and control

- **Macrolides** (azithromycin, clarithromycin) or **fluoroquinolones** (ciprofloxacin, levofloxacin) is the treatment of choice.
- **! β -lactam antibiotics are ineffective. !**
- Prevention requires identification of the bacterial source.
- Reducing number of organisms in water supply is adequate control.
- Hyperchlorination, superheating and copper-silver ionization of the water supply can eliminate bacteria.



References

- Mahon C.R., Lehman D.C., Manuselis G. (2023) Textbook of diagnostic microbiology, 7th edition. Elsevier Inc., Philadelphia.
- Carroll K.C., Pfaller M.A., Karlowasy J.A., et al. (2023) Manual of clinical microbiology, 13th edition. ASM Press, Washington DC.
- Murray P.R., Rosenthal K.S. and Pfaller M.A. (2020) Medical microbiology, 9th edition. Elsevier Inc., Philadelphia.